CS60092: INFORMATION RETRIEVAL

Programming Assignment 3
Learning to Rank

https://github.com/trishitasaha/Outliers_IR_A3.git

Submitted By

Trishita Saha (26CS91R13)

Adrija Guha (26CS71P02)

K. Tharun Selvam (22CS30033)

Tirthankar Sardar (26CS60R06)

Ditsa Kundu (26CS60R01)

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Department of Computer Science and Engineering

Indian Institute of Technology Kharagpur

Kharagpur, West Bengal, India

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1. Problem Statement

The objective of this assignment is to apply techniques to rank documents according to their relevance to a given query. The experiments are conducted using the Cranfield document collection and its associated human relevance judgments.

Let Q denote the set of queries and D the set of documents. For a query $q \in Q$, the task is to produce an ordered list of documents such that documents with higher relevance are ranked above those with lower relevance.

Let $y(q, d)$ denote the relevance judgment associated with query-document pair (q, d). The goal is to learn a ranking function

$$f(q, d; \boldsymbol{\theta})$$

parameterised by a set of learnable weights $\boldsymbol{\theta}$, such that the ranking score reflects the relevance of document d to query q.

For two documents d_i and d_j associated with the same query, the desired ordering is

$$y(q, d_i) > y(q, d_j) \implies f(q, d_i; \boldsymbol{\theta}) > f(q, d_j; \boldsymbol{\theta}).$$

The assignment investigates two approaches to learning such a ranking function.

Method A: Boolean Weighted Zone Scoring

Each document is divided into four fields: title, author, bibliography and text body. A Boolean matching function indicates whether a query matches each field.

Let the corresponding Boolean matching vector be

$$\mathbf{b}(q, d) = [b_t, b_a, b_b, b_x]^T,$$

where the components represent title, author, bibliography and text-body matches, respectively.

The weighted ranking score is

$$S_A(q, d) = w_t b_t + w_a b_a + w_b b_b + w_x b_x.$$

The objective is to learn the field weights $\mathbf{w} = [w_t, w_a, w_b, w_x]^T$ using the relevance judgments, and rank documents in descending order of their scores.

Method B: Learning Real-Valued Weights Using SVM

For each query-document pair, a 22-dimensional feature vector is constructed:

$$\mathbf{x}(q,d) = [x_1(q,d), x_2(q,d), \ldots, x_{22}(q,d)]^T.$$

A linear ranking function is learned:

$$S_B(q,d) = \mathbf{w}^T \mathbf{x}(q,d).$$

The SVM learns real-valued feature weights from pairwise document preferences derived from relevance judgments. The resulting scores are used to rank documents for each query.

The performance of both methods is evaluated using Precision, Recall and F1 score, with NDCG additionally used to assess graded ranking quality.

Dataset and Relevance Judgments

The experiments use the Cranfield text collection, which contains documents, queries and human relevance assessments. The original relevance judgments were mapped to numerical values as listed in Table 1:

Table 1: Relevance label mapping

Original Label	Mapped Relevance
1	4
2	3
3	2
4	1
-1	0
Unjudged pair	0

Higher mapped values indicate greater relevance. Unjudged pairs were assigned relevance zero for this experiment. This is an experimental assumption.

2. Implementation Details

The following flowchart illustrates the overall implementation workflow, highlighting the different methods and steps involved in the project.

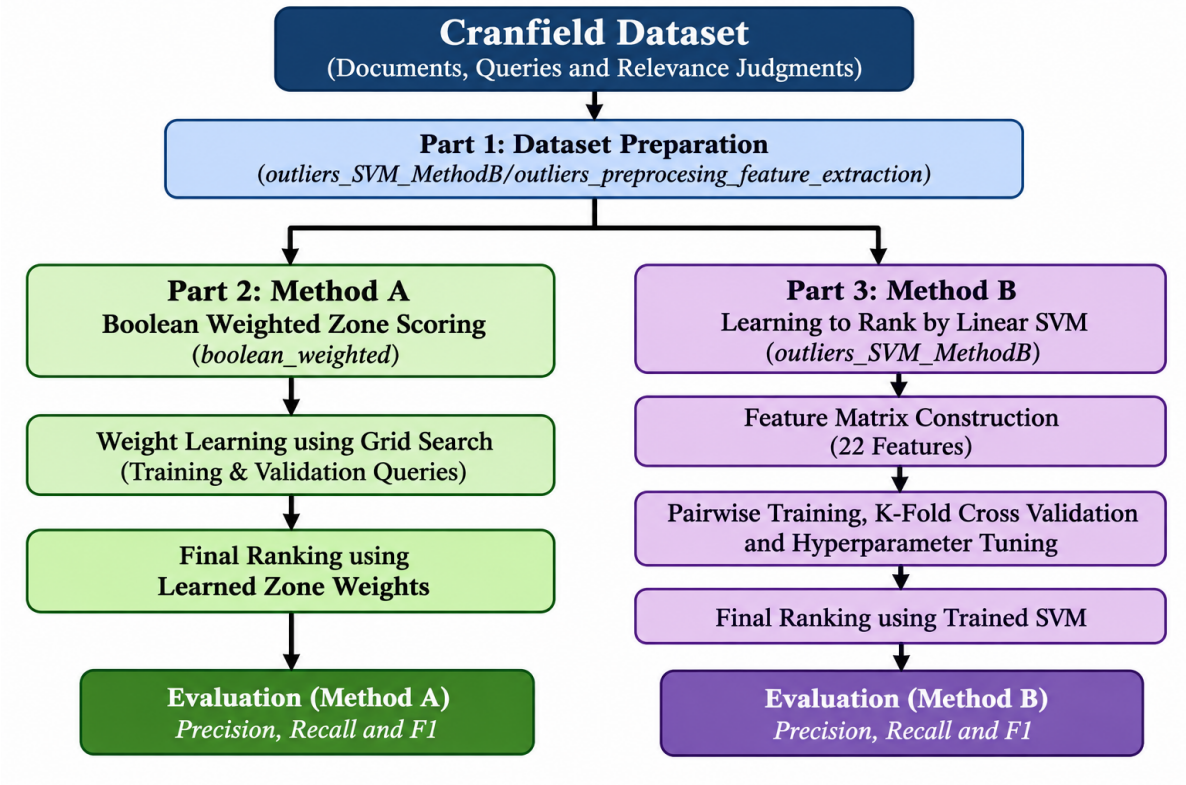


Figure 1: Overall implementation workflow of the system.

2.1 Method A: Boolean Weighted Zone Scoring

The Boolean Weighted Zone Scoring method ranks documents by considering the occurrence of query terms in four document fields: title, author, bibliography, and body. Each field is independently assigned a Boolean matching score, and the contribution of each field to the final ranking is determined by its corresponding weight.

2.1.1 Boolean Matching

For a given query q and document d, the query terms and document field terms are converted into lowercase sets to eliminate duplicate terms. For each field f, the number of distinct matching query terms is calculated using set intersection:

$$M_f(q, d) = |Q(q) \cap F_f(d)|$$

A Boolean matching score is assigned according to a minimum-match threshold t:

$$B_f(q,d) = \begin{cases} 1, & \text{if } M_f(q,d) \geq t, \\ 0, & \text{otherwise.} \end{cases}$$

The original OR-based matching corresponds to a threshold of $t = 1$. To investigate the effect of requiring stronger matches, thresholds of 1, 2, and 3 were considered during experimentation.

2.1.2 Weighted Zone Scoring

The Boolean scores from the four fields are combined using a weighted sum:

$$S_A(q,d) = w_t B_t(q,d) + w_a B_a(q,d) + w_b B_b(q,d) + w_x B_x(q,d)$$

where w_t, w_a, w_b, and w_x represent the weights assigned to the title, author, bibliography, and body fields, respectively.

Only documents having a positive final score are considered for retrieval. The retrieved documents are sorted in descending order of their weighted scores. When multiple documents have identical scores, ascending document ID is used as the tie-breaking criterion.

2.1.3 Learning Field Weights

The field weights are learned through an exhaustive grid search. Each field weight is selected from the integer range $w_f \in 0, 1, 2, 3, 4, 5$. The all-zero combination is excluded, resulting in $6^4 - 1 = 1295$ possible weight combinations for each matching threshold.

For each threshold, the weight combination producing the highest training F1-score at a cutoff of 10 is selected. The resulting configurations are evaluated on the validation queries, and the configuration with the highest validation F1-score is chosen as the final model.

The selected weights and matching threshold are subsequently kept fixed while evaluating the model on the held-out test queries. This ensures that the test relevance judgments are not used during weight learning or configuration selection.

2.2 Method B: Pairwise Linear SVM

2.2.1 Feature Representation

Each query-document pair is represented by the following 22 features as listed in Table 2.

Table 2: The 22 query-document features used by the Ranking SVM

No.	Feature	No.	Feature
1	Query term number	12	Normalized TF maximum
2	Query term ratio	13	Normalized TF mean
3	Stream length	14	Normalized TF variance
4	IDF	15	TF-IDF sum
5	TF sum	16	TF-IDF minimum
6	TF minimum	17	TF-IDF maximum
7	TF maximum	18	TF-IDF mean
8	TF mean	19	TF-IDF variance
9	TF variance	20	Boolean model score
10	Normalized TF sum	21	Vector Space Model score
11	Normalized TF minimum	22	BM25 score

These features capture query-document matching, term-frequency statistics, term weighting and traditional retrieval-model scores.

2.2.2 Pairwise Ranking Formulation

A pairwise approach was used to transform the ranking problem into a binary classification problem.

For each query, document pairs having different relevance labels were generated. If document d_i has a higher relevance label than document d_j, their feature difference is

$$\mathbf{z}_{ij} = \mathbf{x}(q, d_i) - \mathbf{x}(q, d_j).$$

The pair is assigned a positive preference label. The reverse ordering corresponds to a negative label.

The linear SVM learns a weight vector that separates these pairwise preferences. Its squared-hinge optimisation objective is

$$\min_{\mathbf{w}} \frac{1}{2}\|\mathbf{w}\|_2^2 + C \sum_{(i,j)} \max(0, 1 - \mathbf{w}^T \mathbf{z}_{ij})^2.$$

Here, C controls the trade-off between model regularisation and pairwise constraint violations.

Feature normalisation was performed using `StandardScaler`. A linear SVM classifier (`LinearSVC`) with L2 regularisation and squared-hinge loss was trained using pairwise feature

differences. The trained model assigns a real-valued score to each query-document pair, given by $S_B(q, d) = \mathbf{w}^T \mathbf{x}(q, d)$.

For each query, documents are sorted in descending order of these scores to obtain the final ranked list. The saved model and scaler can be reused to rank documents for unseen queries without requiring relevance judgments at prediction time.

3. Plan of Experiments

The experiments were designed to learn ranking weights, evaluate retrieval performance and compare the two approaches against traditional retrieval baselines.

3.1 Method A: Boolean Weighted Zone Scoring

The experiments were designed to determine the contribution of individual document fields to retrieval performance and to identify an appropriate minimum-match threshold.

The following experimental procedure was adopted:

1. **Boolean Matching Threshold Selection:**

   Three minimum-match thresholds ($t = 1, 2, 3$) were considered. A threshold of 1 represents the conventional OR-based matching strategy, whereas higher thresholds require more distinct query terms to match within a field.

2. **Field Weight Grid Search:**

   For each threshold, an exhaustive search was performed over all non-zero combinations of four field weights. Each weight was selected from the range $\{0, 1, 2, 3, 4, 5\}$, resulting in $6^4 - 1 = 1295$ combinations.

3. **Training Phase:**

   For every weight combination, documents were ranked using their weighted Boolean scores. Precision, Recall, and F1-score were calculated at a cutoff of 10. The combination achieving the highest training F1-score was retained for each threshold.

4. **Validation Phase:**

   The best weight configuration obtained for each threshold was evaluated on the validation queries. The threshold and corresponding weights producing the highest validation F1-score were selected as the final configuration.

5. **Test Evaluation:**

   After selecting the configuration using validation performance, the weights and threshold were frozen. The final configuration was evaluated on the held-out test queries to measure its retrieval effectiveness.

Precision, Recall, and F1-score were used as the evaluation metrics. The experiments focused on the top 10 retrieved documents for each query.

This experimental procedure allows the effect of different field combinations and matching thresholds to be studied systematically while keeping the test set separate from the weight-learning process.

Extended Experiment of Boolean Weighted Zone Scoring:

The soft experiment tested whether ranking documents by how much of a query appears in each document field gives more useful ordering than the Boolean model's all-or-nothing field matches. For each query and field (‘title‘, ‘author‘, ‘bibliography‘, and ‘body‘), it computes the proportion of distinct query terms present. The four proportions are combined with learned field weights, and documents with a positive score are ranked by that weighted sum.

The grid search tried non-zero weight combinations from 0 to 5 and selected weights using training-query F1 at the top-10 cutoff. Validation queries were then used to assess the selected configuration. The best weights were title=4, author=0, bibliography=0, and body=3 (normalized to approximately 0.571 title and 0.429 body). This result indicates that, for this dataset and setup, title and body coverage contributed to ranking, while author and bibliography coverage did not improve the selected configuration.

With those weights fixed, the separate test run evaluated 34 queries at cutoff 10. It achieved **mean precision@10 of 0.20294, recall@10 of 0.32981, and F1@10 of 0.22577.** No test query returned fewer than 10 results. These figures describe this dataset split and scoring setup; they are not evidence that the same field weights will generalize to other collections.

3.2 Method B: Pairwise Linear SVM

1. **Initial Train-Test Split:**

   In the initial implementation, the Pairwise Linear SVM was trained using an 80:20 query-level train-test split. The training set was used to generate pairwise preference examples

and fit the model, while the held-out test set was reserved for final evaluation. Initially, the regularization parameter C was selected without performing cross-validation.

2. **Incorporating Cross-Validation:**

   To make hyperparameter selection more reliable, we subsequently incorporated 5-fold cross-validation on the training queries. The training queries were divided into five folds using **GroupKFold**, ensuring that documents belonging to the same query were not distributed across different folds.

3. **Hyperparameter Search:**

   We experimented with the following values of the regularization parameter:

   $C \in \{0.0001, 0.0003, 0.001, 0.003, 0.01, 0.03, 0.1, 1, 10, 100\}$.

4. **Model Selection:**

   Mean NDCG@10 was used as the selection criterion. As shown in Table 3, $C = 0.0003$ produced the highest observed mean validation NDCG@10 and was therefore selected for final training.

Table 3: Five-fold cross-validation results for different values of C (mean and standard deviation)

C	**Precision**		**Recall**		**F1-score**		**NDCG@10**		**Pairwise Accuracy**	
	Mean	Std.	Mean	Std.	Mean	Std.	Mean	Std.	Mean	Std.
0.0001	0.2333	0.0415	0.3944	0.0555	0.2627	0.0401	0.3503	0.0387	0.9048	0.0166
0.0003	**0.2344**	0.0346	**0.3978**	0.0475	**0.2647**	0.0330	**0.3537**	0.0393	0.9050	0.0166
0.001	0.2333	0.0316	0.3936	0.0451	0.2630	0.0309	0.3514	0.0394	**0.9050**	0.0164
0.003	0.2317	0.0305	0.3929	0.0442	0.2617	**0.0303**	0.3490	0.0408	**0.9050**	0.0163
0.01	0.2317	**0.0295**	0.3932	**0.0446**	0.2619	**0.0301**	0.3493	**0.0363**	0.9049	**0.0152**
0.03	0.2317	0.0318	0.3926	0.0472	0.2616	0.0321	0.3491	0.0381	0.9048	0.0161
0.1	0.2317	0.0318	0.3926	0.0472	0.2616	0.0321	0.3488	0.0380	0.9047	0.0161
1	0.2317	0.0318	0.3926	0.0472	0.2616	0.0321	0.3489	0.0381	0.9045	0.0161
10	0.2317	0.0318	0.3926	0.0472	0.2616	0.0321	0.3490	0.0383	0.9045	0.0161
100	0.2317	0.0318	0.3926	0.0472	0.2616	0.0321	0.3489	0.0381	0.9045	0.0161

The selected regularisation parameter was

$$\boxed{C = 0.0003}$$

4. Results

Evaluation Metrics

Precision and Recall measure retrieval effectiveness using binary relevance, where a document is considered relevant if its mapped relevance label is greater than zero.

$$P@K = \frac{\text{Relevant documents in top } K}{K}$$

$$R@K = \frac{\text{Relevant documents in top } K}{\text{Total relevant documents}}$$

The F1 score is the harmonic mean of Precision and Recall:

$$F1@K = \frac{2(P@K)(R@K)}{P@K + R@K}.$$

NDCG evaluates ranking quality using graded relevance:

$$DCG@K = \sum_{i=1}^{K} \frac{2^{rel_i} - 1}{\log_2(i+1)}$$

$$NDCG@K = \frac{DCG@K}{IDCG@K}.$$

4.1 Method A Results: Boolean Weighted Zone Scoring

The grid-search experiment evaluated 1295 field-weight combinations for each of the three Boolean matching thresholds. The best configuration for each threshold was selected using training F1-score, followed by validation-based configuration selection.

The final configuration selected using validation performance was:

Learned Field Weights

The learned field weights and their normalized values are presented in Table 4.

The resulting ranking function is:

$$S_A(q,d) = B_t(q,d) + B_x(q,d) \tag{1}$$

Thus, a document receives one point for matching at least three distinct query terms in its

Table 4: Learned Field Weights for Boolean Weighted Zone Scoring

Document Field	Learned Weight	Normalized Weight
Title	1	0.50
Author	0	0.00
Bibliography	0	0.00
Body	1	0.50

title and another point for satisfying the same condition in its body. Matches in the author and bibliography fields do not contribute to the final score.

Performance of the Selected Configuration

The selected configuration uses a minimum match threshold of 1 and field weights `title`=1, `author`=0, `bibliography`=0, and `body`=1. Its performance across the training, validation, and test stages is presented in Table 5.

Table 5: Performance of the Selected Boolean Weighted Zone Scoring Configuration

Evaluation Stage	F1-score
Training	0.24767
Validation	0.21457
Test	0.23713

The final test-set retrieval metrics at cutoff 10 are shown in Table 6. Precision@10 uses a fixed denominator of 10.

Table 6: Test-Set Retrieval Performance at Cutoff 10

Metric	Test Score
Precision@10	0.20882
Recall@10	0.32140
F1-score@10	0.23713

The selected configuration achieved a test F1-score of 0.23713, with Precision@10 of 0.20882 and Recall@10 of 0.32140. The test F1-score is higher than the validation F1-score of 0.21457.

4.2 Method B Results: Pairwise Linear SVM

Baseline Comparison

The final SVM model was compared with BM25 and Vector Space Model (VSM) on the same 45 held-out queries. The evaluation was performed at $K = 5$ and $K = 10$, using Precision, Recall, F1-score and NDCG. The comparative results are presented in Table 7.

Table 7: Comparison of Ranking SVM, BM25 and VSM on the test set

Method	K	Precision	Recall	F1-score	NDCG
Ranking SVM	5	0.3733	0.3542	0.3243	0.3974
BM25	5	0.3733	**0.3681**	**0.3348**	**0.4175**
VSM	5	**0.3911**	0.3600	0.3344	0.3930
Ranking SVM	10	**0.2756**	**0.4692**	**0.3124**	0.4329
BM25	10	**0.2756**	0.4571	0.3094	**0.4488**
VSM	10	0.2644	0.4623	0.3033	0.4066

At $K = 5$, VSM achieved the highest precision of 0.3911. BM25 achieved the highest recall and F1 score, along with the highest NDCG of 0.4175. Ranking SVM obtained an NDCG of 0.3974. At $K = 10$, Ranking SVM achieved the highest recall (0.4692) and F1 score (0.3124). Its precision was equal to BM25 at 0.2756. BM25 achieved the highest NDCG@10 of 0.4488.

Feature Importance

To understand the contribution of individual features to the learned ranking model, we analysed the absolute magnitudes of the standardized SVM coefficients. The importance percentage was calculated by dividing the absolute coefficient of each feature by the sum of the absolute coefficients across all 22 features. The top 10 features are presented in Table 8.

Table 8: Top 10 features ranked by standardized coefficient magnitude

Rank	Feature	Importance (%)
1	Normalized TF sum	10.89
2	TF-IDF maximum	9.92
3	BM25 score	9.87
4	TF-IDF mean	9.28
5	Query term ratio	8.86
6	Query term number	7.20
7	TF mean	6.83
8	TF sum	6.81
9	Stream length	6.49
10	TF-IDF variance	5.18

5. Discussion of Results

5.1 Method A Discussion

5.1.1 Discussion of Experimental Results

The experimental results show that the document fields and minimum-match threshold affect Boolean Weighted Zone Scoring performance.

The grid search selected a minimum-match threshold of 1, with weights of 1 for the title and body fields and weights of 0 for the author and bibliography fields. Under this threshold, a field receives a Boolean match when it contains at least one distinct query term. The selected configuration achieved the highest validation F1-score among the configurations evaluated.

5.1.2 Interpretation of Learned Field Weights

The title and body fields received equal weights in the selected scoring function. A match in either field contributes 1 to the Boolean score. When documents have the same Boolean score, the ranking uses the weighted number of distinct query-term matches as a tie-breaker, followed by document ID.

The author and bibliography fields received zero weights in this run. This means their matches did not contribute to the selected ranking function. It does not establish that these fields are universally irrelevant; their usefulness may depend on the query and document collection.

5.1.3 Interpretation of Evaluation Results

At a cutoff of 10, the selected configuration achieved Precision@10 of 0.20882, Recall@10 of 0.32140, and an F1-score of 0.23713 on the test set. Recall was higher than precision, indicating that the system retrieved a greater share of relevant documents than the share of retrieved documents judged relevant.

The training, validation, and test F1-scores were 0.24767, 0.21457, and 0.23713, respectively. Test F1 was higher than validation F1 and lower than training F1. These results show a difference in performance across the splits; the test result alone does not establish how the system will generalize to other query sets.

5.2 Method B Discussion

5.2.1 Comparison with Baseline Methods

The experiments show that learning a ranking function from relevance preferences is useful, but does not automatically guarantee better performance than traditional retrieval models. BM25 remained strong in NDCG, possibly because its carefully designed term-frequency saturation and document-length normalisation provide a robust ranking signal for the Cranfield collection.

5.2.2 Statistical Significance Testing

Paired Wilcoxon signed-rank tests were performed to compare Ranking SVM with BM25 using per-query NDCG values.

Table 9: Wilcoxon signed-rank test: SVM versus BM25

Metric	Mean Difference	p-value
NDCG@5	-0.0200	0.8130
NDCG@10	-0.0159	0.6487

The mean difference is calculated as SVM minus BM25. Both p-values are greater than 0.05. Therefore, the tests do not establish a statistically significant difference between the two methods. Although BM25 achieved higher mean NDCG at both cut-offs, the observed differences should not be considered statistically conclusive.

5.2.3 Feature Importance Analysis

As by Table 8 Normalized TF sum received the highest relative importance, followed by TF-IDF maximum and BM25. This indicates that the model combines term-frequency information, term weighting and an established retrieval-model score.

The distribution of importance across several features suggests that the ranking function uses multiple retrieval signals rather than depending entirely on a single feature.

Coefficient importance represents the model's learned linear contribution and should not be interpreted as causal influence. Correlations between features may also affect coefficient magnitudes.

Limitations

The experiments have some limitations. Unjudged documents were treated as non-relevant, although the absence of a relevance judgment does not necessarily mean that a document is irrelevant. This can introduce noise into both pairwise training and evaluation. Furthermore, the model was evaluated on a single query-level split of the Cranfield collection, so its performance on other document collections remains uncertain.

5.3 Comparative Analysis

The Pairwise Linear SVM approach achieved a Precision@10 of 0.2756, Recall@10 of 0.4692, and F1-score@10 of 0.3124. These values are higher than the corresponding Boolean Weighted Zone Scoring results.

The SVM approach also achieved an NDCG@10 of 0.4329, demonstrating its ability to incorporate graded relevance information into document ranking. The Boolean method, in contrast, uses binary field matching signals and does not directly model the degree of relevance of individual documents.

Traditional retrieval methods also performed better than the Boolean method in the reported SVM comparison. BM25 achieved an F1-score@10 of 0.3094 and NDCG@10 of 0.4488, while VSM achieved an F1-score@10 of 0.3033 and NDCG@10 of 0.4066.

The lower performance of Boolean Weighted Zone Scoring can be attributed to its relatively simple representation of query-document matching. Since each field contributes only a binary value, the model does not distinguish between documents having different numbers of matching terms once the minimum threshold has been satisfied. It also does not incorporate term-frequency information, inverse document frequency, or document-length normalization.

Overall, Boolean Weighted Zone Scoring provides a simple and interpretable approach to document ranking. Its learned weights identify the most useful fields for the evaluated configuration. However, its limited matching representation makes it less effective than the Pairwise Linear SVM and traditional retrieval approaches in the reported experiments.

6. Conclusion

This assignment investigates two approaches to Learning-to-Rank on the Cranfield collection: Boolean Weighted Zone Scoring and Pairwise Linear SVM.

The Boolean Weighted Zone Scoring method learns field weights through an exhaustive grid

search over the title, author, bibliography, and body fields. The selected configuration assigned weights of 1 to the title and body and 0 to the author and bibliography, with a minimum matching threshold of 1. On the held-out test set, it achieved **Precision@10 of 0.20882, Recall@10 of 0.32140, and F1@10 of 0.23713.** Documents with the same Boolean score are further ordered by their weighted number of distinct query-term matches. These results show that field selection and matching thresholds affect retrieval performance, while binary field scores can still produce ties between documents.

The SVM approach learns a real-valued linear ranking function from 22 query-document features and pairwise preferences derived from relevance judgments. It was evaluated against BM25 and VSM on 45 held-out queries. The final SVM model achieved **Precision@10 of 0.2756 Recall@10 of 0.4692 and F1@10 of 0.3124 and NDCG@10 of 0.4329**. Statistical testing did not establish a significant difference between Ranking SVM and BM25.

Overall, the assignment demonstrates how supervised learning can be applied to combine document and query features to produce effective ranked retrieval results.

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